

The resolvent of C_1 and C_2 is $\neg P$ which is the logical consequence of C_1 and C_2 .

Proof: Consider two clauses C_1 and C_2 .

C_1 contains L and C_2 contains $\neg L$.

Writing C_1 as $L \vee C_1'$

and C_2 as $\neg L \vee C_2'$

resolvent of C_1 and C_2 is $C = C_1' \vee C_2'$

Now we want to show C is the logical consequence of C_1 and C_2 . In essence, we want to show that if C_1 and C_2 are true, C is true.

Let C_1 and C_2 be both True.

$\therefore (L \vee C_1')$ is True and $(\neg L \vee C_2')$ is True.

By defn. of OR, at least one of L or C_1' is True. (Case 1)

If C_1' is True, By defn. of OR, $C_1' \vee C_2'$ is True.

If L is True, By defn. of negation, $\neg L$ is False. Since $(\neg L \vee C_2')$ is True, by defn. of OR, C_2' is True. By defn. of OR, $(C_1' \vee C_2')$ is True.

By defn. of OR, at least one of $\neg L$ or C_2' is True. (Case 2)

If C_2' is True, by defn. of OR, $(C_1' \vee C_2')$ is True.

If $\neg L$ is True, by defn. of negation, $\neg(\neg L)$ is False. By double negation law, L is False. Since $(L \vee C_1')$ is True, by defn. of OR, C_1' is True.

By defn. of OR, $(C_1' \vee C_2')$ is True.

$\therefore C_1 \wedge C_2 \Rightarrow C_1' \vee C_2'$ or, $C_1 \wedge C_2 \Rightarrow C$. C is the logical consequence of C_1 and C_2 . $\square$

The Resolution Principle: Given a set S of clauses, a (resolution) deduction of C from S is a finite sequence $C_1, C_2, \dots, C_k$ of clauses such that each C_i is either a clause in S or a resolvent of clauses preceding C and $C_k = C$. A deduction of $\square$ (empty clause) is called a refutation or a proof of S .